

Alok Kumar Gupta

Gorakhpur, Uttar Pradesh, India, 273306

(+91) 9305628738 | work.alokgupta@gmail.com | linkedin.com/in/trooperr | github.com/troooperr

PROFESSIONAL SUMMARY

Full Stack Developer with experience building scalable software systems, real-time applications, and production-grade web platforms. Strong foundation in data structures, algorithms, distributed systems, REST APIs, WebSockets, database optimization & concurrency.

EDUCATION

Pranveer Singh Institute of Technology

B.Tech CSE, 77.56% (till 6th sem)

2023 - Pursuing

Kanpur

Red Rose Public School

Senior Secondary Education, 81.20%

2022

Lucknow

Red Rose Public School

Secondary Education, 92.83%

2020

Lucknow

TECHNICAL SKILLS

Programming Languages: C, C++, JavaScript, TypeScript

Frontend Technologies: React.js, HTML5, CSS3, Tailwind CSS, GSAP, Three.js

Backend Technologies: Node.js, Express.js, REST APIs, WebSockets, Socket.IO

Databases & ORM: MongoDB, PostgreSQL, Prisma

Testing & Tools: Jest, Supertest, Artillery, Git, GitHub, Docker, Postman

Core Concepts: Data Structures and Algorithms, Object Oriented Programming, Database Management System

Authentication & Security: JSON Web Tokens (JWT), OAuth 2.0, Bcrypt, Role-Based Access Control (RBAC)

PROJECTS

- Restaurant POS System (GitHub | Live) | React, Node.js, Express.js, MongoDB, Redis** **Feb 2026 - June 2026**
 - Built a production-grade **Restaurant POS & Inventory Management System** supporting **100+ active table sessions**, **500+ orders/hour**, real-time billing, KOT generation, inventory management, and role-based operations.
 - Designed a **parent-child inventory engine** managing **1,000+ SKUs**; optimized stock synchronization using MongoDB bulk operations (**500+ updates/batch**) and achieved **< 15 ms** session lookups through Redis caching and database indexing.
 - Engineered secure **REST APIs** with JWT authentication, RBAC, rate limiting, and an RSA-based thermal printer gateway, supporting **10,000+ daily API requests** while preventing duplicate billing under concurrent workloads.
 - Deployed in a live restaurant and validated production readiness through **Artillery load testing**, **Jest/Supertest API testing**, automated cron jobs, health monitoring, and real-time synchronization across **150+ connected devices**.
- ShramSetu - Job Allocation Platform (GitHub | Live) | MERN, WebSockets, Geospatial Indexing** **Oct 2025 - Jan 2026**
 - Built a **real-time job allocation platform** supporting **100+ worker and employer accounts**, enabling instant job matching, role-based workflows, live job updates, location-aware worker discovery, and seamless client-worker communication.
 - Developed secure **REST APIs** with JWT authentication and RBAC, managing **20+ API endpoints** for authentication, job posting, applications, and user management.
 - Optimized **MongoDB geospatial indexing**, reducing location-based job assignment latency by **40%**.
 - Implemented **WebSocket-based real-time communication** for live location tracking, job status updates, and event-driven notifications, supporting **50+ concurrent client connections**.
- iPhone15 Website Clone (GitHub | Live) | React.js, Three.js, GSAP, Tailwind CSS** **Oct 2024 - Dec 2024**
 - Built a pixel-perfect iPhone15 product showcase with responsive layouts, smooth animations, interactive 3D models, and cross-device compatibility.
 - Developed reusable **React components** with modular architecture, responsive layouts, and multiple product variants.
 - Implemented **GSAP scroll animations**, smooth transitions, and interactive effects for an engaging user experience.
 - Integrated optimized **Three.js 3D models** for interactive product visualization with efficient rendering performance.

CODING PLATFORMS

- LeetCode:** Solved 450+ problems, achieved 1707 rating, ranked Top 12% globally, strengthening problem-solving skills.
- CodeChef:** Competed in multiple contests, earned 1-Star, peak 1367 rating, demonstrating competitive programming skills.
- HackerRank:** Achieved 5-Star in C++, completed advanced exercises and challenges, strengthening coding expertise.

ACHIEVEMENTS & CERTIFICATIONS

- Byte-Battle Hackathon Finalist:** Finalist among 50+ teams for developing a scalable full-stack web applications.
- Completed **Introduction to User Experience Design** from Georgia Institute of Technology, gaining foundational UX skills.